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Lecture 7: Probabilistic Prime Number Tests

Little Fermat's Theorem:

$$p \in \mathbb{P}, a \in \mathbb{N}, (a,p) = 1 => \ a^{p-1} = 1(p)$$

Question: $n \in \mathbb{N}, a \in \mathbb{N}\ with\ a^{n-1} = 1(n) => n \overset{?}{\in} \mathbb{P}$

$$2^{n-1} = 1(n)$$
$$n = 2, 3, \ldots$$

n=341 – the number is not prime, but it pretends to be prime
⇨ Smallest pseudoprime number to base 2

Definition: Let m be a composite number with $(a,m) = 1$ and $a^{m-1} = 1(m)$ or $a^m = a(m)$, then m is called pseudoprime to base a

n=341 – the number is not prime, but it pretends to be prime

- n=341 is smallest pseudoprime number to base 2
- n=341 is not a pseudoprime to base 3

In other words: Pseudoprime numbers satisfy Fermat's little theorem even though they are not prime

Existence of composite numbers that are pseudoprime to all (coprime) bases:

A composite number is called a Charmichael number iff $a^{m-1} = 1(m)$
or $a^m = 1(m)$ applies to all bases with $(a,m) = 1$

- m=561 is the smallest Carmichael number
- Carmichael numbers are free of squares
- Factoring Carmichael numbers contains at least 3 different prime factors
- Large Carmichael numbers are almost always classified as prime ☹
- There are infinitely many Carmichael numbers: 561, 1105, 1729, 2465, 2821, ...

Theorem: For $m \in \mathbb{N}, m \geq 3$, let

$$F_m = \{a \in \mathbb{Z}_m \mid a^{m-1} = 1(m)\}$$

be the set of bases for which m passes the Fermat test.

If m is not a prime number, then F_m contains the bases that "fool" the Fermat test.

Let $m \in \mathbb{N}, m \geq 3$, be a composite and not a Carmichael number, then the following applies:

$$|F_m| \leq \frac{\mathbb{Z}_m}{2}$$

alg **notPrime** $(\boldsymbol{k} \in \boldsymbol{U}_+, \boldsymbol{k} \geq \mathbf{3})$

    Randomly pick an $\boldsymbol{a} \in (\mathbf{1}, \dots, \boldsymbol{k} - \mathbf{1})$ **with** $(\boldsymbol{a}, \boldsymbol{k}) = \mathbf{1}$

    If $\boldsymbol{a}^{\boldsymbol{k}-\mathbf{1}} \neq \mathbf{1}(\boldsymbol{k})$
        then Output: $\boldsymbol{k}$ ***is not prime***
        otherwise Output: $\boldsymbol{k}$ ***prime***?
endalg

$\overline{\mathbb{P}}$ = COMPOSITES $\in$ RPP

(1) $Prob[k \in \overline{\mathbb{P}} \mid k \notin \overline{\mathbb{P}}] = 0$

(2) $Prob[k \notin \overline{\mathbb{P}} \mid k \in \overline{\mathbb{P}}] \leq \frac{1}{2}$

(3) notPrime is polynomial

The probability of error is therefore at most $\frac{1}{2}$. If the algorithm is now carried out for *l* rounds in which the base *a* is chosen anew at random and independently, then the probability of error is at most $\frac{1}{2^l}$; so, it can be made as small as you want.

➔ Executing the algorithm l times leads to an error probability of $\leq \frac{1}{2^l}$

Miller Rabin Algorithm and b-Sequences

$$\mathbb{Z}_m \; is\ a\ field \text{ iff } \; m \in \mathbb{P}$$

$$x^2 = 1$$

$$\mathbb{Z}_m: x = 1, x = -1 \;\; => trivial\ solution$$

$x \in \mathbb{Z}_m$ is called a non-trivial square root of 1 modulo m if $x^2 = 1$ and $x \neq 1$ and $x \neq -1$.

e.g., $m = 15$: $x = 4, x = -4 \;\; => non - trivial\ solution$

In $\mathbb{Z}_m$, x^2 has the solution $x = \pm 1$ iff $m \in \mathbb{P}$

In other words: If there is a nontrivial square root modulo m, then m is a composite number.

$$m \in \mathbb{N}$$

$$s = max\{\, r \in \mathbb{N} \mid 2^r \mid m - 1\}$$

$$d = \frac{m-1}{2^s}$$

$$b \in \mathbb{Z}_m: b - sequence$$

$$\langle b^{2^0 d}, b^{2^1 d}, b^{2^2 d}, \dots, b^{2^{s-1} d}, b^{2^s d} \rangle \;\; mod \;\; (m)$$

If $m \in \mathbb{P}$:

$$b^{2^s d} = b^{2^s * \frac{m-1}{2^s}} = b^{m-1} = 1(m)$$

Theorem: *Let be* $p \in \mathbb{P},\; s = max\{r \mid 2^r \mid p - 1\},\; d = \frac{p-1}{2^s}, b \in \mathbb{N}$ *with* $(b, p) = 1$

Then (1) $b^d = 1\ (p)$ or

(2) $\exists r \in \{0, 1, \dots, s - 1\}: b^{2^r d} = -1\ (p)$

If $p \in \mathbb{P}$, then the b-sequence has one of the following forms:

(1) $\langle 1,\ 1, \dots, 1 \rangle$

(2) $\langle -1,\ 1, \dots,\ 1 \rangle$

(3) $\langle ?,\ ?,\ \dots,\ ?, -1, 1, \dots, 1 \rangle$

Last element of the b-sequence: $b^{p-1} = 1\ (p)$

Use the reverse of the b-sequence:

$m \in \mathbb{N}, b \in \mathbb{Z}_m, with\ b - sequence$

(1) $\langle ?,\ ?, \dots ,1,1, \dots ,1\rangle$

(2) $\langle ?, ?, \dots, ?, -1\rangle$

(3) $\langle ?,\ ?,\ \dots\ ,\ ?, ?\rangle$

➔ $m \notin \mathbb{P}$

Definition: $m \in U_+, m \geq 3, m - 1 = 2^s d\ \ with\ \ d \in U_+, b \in \mathbb{Z}_m$

If $b^d = 1(m)$ or $b^{2^r d} = -1(m)$ holds for an $r \in \{0, 1, \dots, s-1\}$, then m is called **strong pseudoprime** to base b.

Theorem: $m \in U_+, m \geq 3, composite$, then the number of bases for which m is **strongly pseudoprime** is at most $\frac{\boldsymbol{m-1}}{\boldsymbol{4}}$

➔ Miller Rabin Algorithm

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algorithm MILLER-RABIN(n ∈ U+, n ≥ 3)
    Compute d and s with n − 1 = d · 2^s and d positive uneven
    Randomly pick an a ∈ {2, 3, . . . , n − 2}
    b := a^d(n)
    if b = 1 (n) or b = −1 (n): Output: n is prime?
    for r := 1 to s − 1 do
        b := b^2 (n)
        if b = −1 (n): Output: n is prime?
        if b = 1 (n): Output: n is not prime
    endfor
    Output: n is prime?
endalgorithm MILLER-RABIN
```

Miller Rabin Algorithm ∈ RPP

(1) $Prob[Output\!: n\ is\ not\ prime \mid n \in\ \mathbb{P}] = 0$

(2) $Prob[Output\!: n\ is\ prime? \mid n \notin\ \mathbb{P}]\ \leq \frac{1}{4}$

(3) $Prob[l - times\ Output\!: n\ is\ prime? \mid n \notin\ \mathbb{P}]\ \leq \frac{1}{4^l}$

(4) $Prob[after\ l\ excecutions\ Output\!: n\ is\ not\ prime \mid n \notin\ \mathbb{P}]\ \geq 1 - \frac{1}{4^l}$

(5) $O\big(l * (\log n)\big)$ arithmetical operations or $O(l * (\log n)^3)$ bit operations